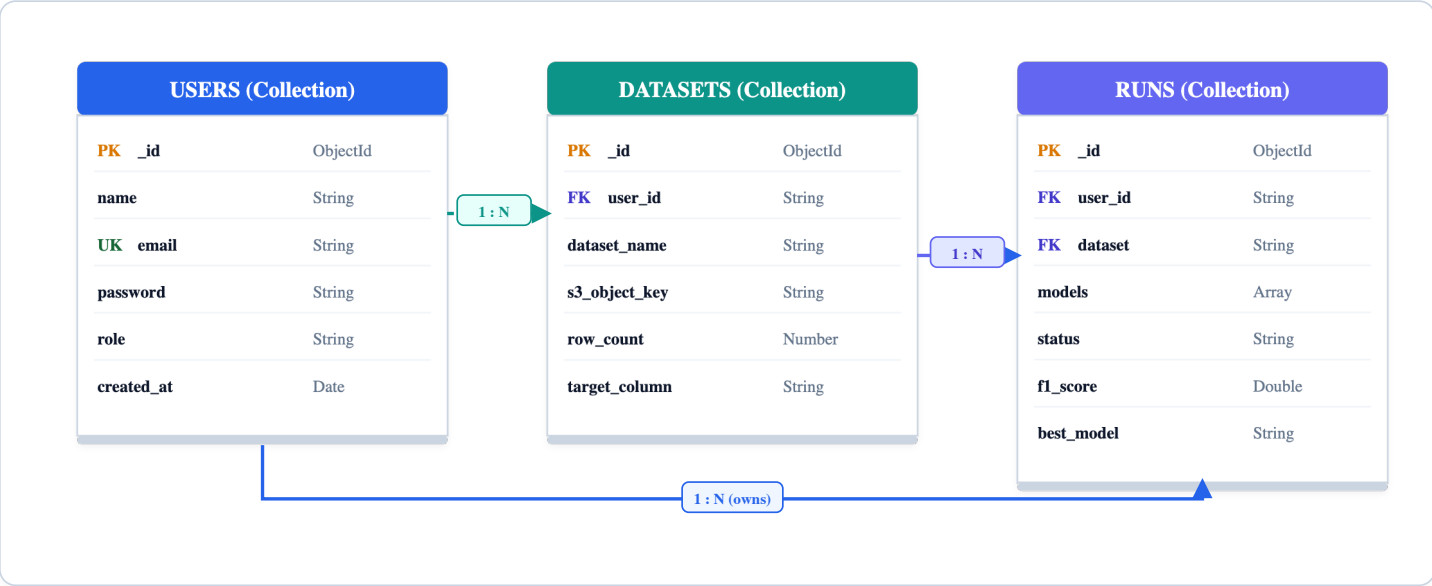


Visual Document Entity Relationship



Detailed MongoDB Collections Data Dictionary

db.usersAuthentication & User Management			
Field Name	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated primary key document ID
name	String	Required	Full display name of operator / researcher
email	String	UK Index	Unique operator login email identifier
password	String	Bcrypt Hash	Bcrypt-hashed password string for JWT auth
role	String	Default: "User"	Access permission role ("User" / "Admin")
created_at	Date	UTC Date	UTC timestamp of account registration

db.runsAutoML Pipeline Runs & Telemetry			
Field Name	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated primary key document ID
user_id	String	FK	Foreign Key linking to users._id
id	String	Logical Key	Unique run key (e.g. run_ibm_active)
dataset	String	FK Ref	Dataset filename processed (e.g. ibm_hr_attrition.csv)
models	Array[String]	Array	List of algorithms executed (Random Forest, XGBoost, LightGBM)
status	String	Enum	State: "Training...", "Completed", "AUTONOMOUS_RETRAINING..."
f1_score	Double	Metric	F1 score of winning model
best_model	String	Algorithm	Top-performing model algorithm selected by AutoML engine
model_scores	Object	Embedded Doc	Dictionary containing accuracy & F1 scores for each trained model

db.datasetsMinIO S3 Dataset Metadata			
Field Name	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated primary key document ID

FIELD NAME	TYPE	CONSTRAINT	DESCRIPTION
user_id	String	FK	Foreign Key referencing users._id
dataset_name	String	Filename	Original dataset filename (e.g. ibm_hr_attrition.csv)
s3_object_key	String	MinIO Key	Full object path in S3 bucket ({user_id}/{filename})
row_count	Number	Count	Total record count in dataset CSV
target_column	String	Target Name	Target classification column name (e.g., "Attrition")